



Low-temperature speciation and chemical kinetic studies of *n*-heptane



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ABSTRACT

Although there have been many ignition studies of *n*-heptane—a primary reference fuel—few studies have provided detailed insights into the low-temperature chemistry of *n*-heptane through direct measurements of intermediate species formed during ignition. Such measurements provide understanding of reaction pathways that form toxic air pollutants and greenhouse gas emissions while also providing key metrics essential to the development of chemical kinetic mechanisms. This paper presents new ignition and speciation data taken at high pressure (9 atm), low temperatures (660–710 K), and a dilution of inert gases-to-molecular oxygen of 5.64 (mole basis). The detailed time-histories of 17 species, including large alkenes, aldehydes, carbon monoxide, and *n*-heptane were quantified using gas chromatography. A detailed chemical kinetic mechanism developed previously for oxidation of *n*-heptane reproduced experimentally observed ignition delay times reasonably well, but predicted levels of some important intermediate chemical species that were significantly different from measured values. Results from recent theoretical studies of low temperature hydrocarbon oxidation reaction rates were used to upgrade the chemical kinetic mechanism for *n*-heptane, leading to much better agreement between experimental and computed intermediate species concentrations. The implications of these results to many other hydrocarbon fuel oxidation mechanisms in the literature are discussed.

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1. Introduction

n-Heptane (*n*-C₇H₁₆) is an important reference compound used to study the chemical kinetics of large *n*-alkanes, which are significant fractions of complex commercial fuels such as gasoline, kerosene, and diesel fuel. The size and structure of *n*-C₇H₁₆ lead to chemical kinetic features that distinguish long-chain *n*-alkanes such as *n*-C₇H₁₆ from other hydrocarbons, and in particular the strong negative-temperature coefficient (NTC) behavior of long-chain *n*-alkanes. As a component of primary reference fuels, *n*-C₇H₁₆ has been studied extensively in shock tubes [1–7], jet-stirred reactors [8,9], rapid compression machines (RCMs) [10–14], premixed laminar flames [15–19], and flow reactors [20]. There have been a smaller number of studies [8,9,13,15,17,20] in which the products of *n*-C₇H₁₆ oxidation—intermediate and final—have been measured experimentally. The level of detail observed and measured in experimental studies of *n*-C₇H₁₆ combustion and the kinetic complexity in the chemical kinetic reaction mechanisms used to interpret those experiments have grown steadily over the

past 20 years. More detailed experiments have motivated better kinetic models and vice versa; recent advances in theoretical chemistry capabilities, such as the *ab initio* studies described later, have also been important.

Nearly 20 years ago, Dagaut et al. [8] studied the oxidation of diluted stoichiometric *n*-C₇H₁₆ mixtures in a jet-stirred reactor at residence times between 0.1 and 2 s, between pressures of 1–40 atm, temperatures of 550–1150 K, and an inert/O₂ ratio between 90 and 180 (mole basis). Using gas chromatography and mass spectrometry, the authors were able to measure 44 different chemical species. They noted that increasing the pressure from 10 to 40 atm increased CO and CO₂ formation, while cyclic ethers such as cis-2-methyl-5-ethyltetrahydrofuran became more prevalent at higher temperatures as the pressure increased. The shift in cyclic ether production was attributed to QOOH formation, which increases with increasing pressure.

Herbinet et al. [9] very recently studied *n*-C₇H₁₆ oxidation in a jet-stirred reactor, focusing on reactions in the low-temperature regime from 500 to 1100 K and measuring concentrations of species important in that temperature range, including cyclic ethers, alkenes, species with two carbonyl groups and ketohydroperoxides. They used a kinetic model to interpret their measurements, with generally good comparisons between computed and experimental results.

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While they did not measure product compositions, one intention of the Cox et al. [14] study was to explore the extent of reaction of $n\text{-C}_7\text{H}_{16}$ during compression in an RCM. The authors used gas chromatography to measure the amount of $n\text{-C}_7\text{H}_{16}$ that reacted during the 22 ms stroke of their RCM as a function of temperature. They supplemented these data with chemiluminescence measurements of the first stage of ignition and found significant amounts of fuel consumption during the compression stroke when the targeted top-dead center (TDC) temperature exceeded 800 K.

In another study carried out nearly 20 years ago, Minetti et al. [13] sampled stoichiometric $n\text{-C}_7\text{H}_{16}$ mixtures with a dilution of inert gases-to- O_2 of 3.76 at a temperature of 667 K and a pressure of 3.4 bar, i.e., $\chi(n\text{-C}_7\text{H}_{16}) = 0.0187$, $\chi(\text{O}_2) = 0.2061$, $\chi(\text{N}_2) = 0.7$, and $\chi(\text{CO}_2) = 0.0752$. Using gas chromatography/mass spectrometry, they identified 36 distinct compounds formed during an ignition delay time of 41 ms, with the first stage of ignition occurring 30 ms after top dead center in their RCM. They compared their experimental data with model predictions based on an early chemical mechanism by Warnatz [21]. The model adequately reproduced ignition delay times, but did not perform well at predicting the time-histories of many of the intermediate species. The mechanism predicted significant consumption of $n\text{-C}_7\text{H}_{16}$ at the first stage of ignition (80%), while experimental measurements were closer to 20%. Minetti et al. [13] concluded that a great deal more mechanism development was needed before reliable species predictions would be possible.

Finally, Ingemarsson et al. [15], Douste et al. [17] and Held et al. [20] measured intermediate species levels in laminar flames and turbulent flow reactors, but these studies concerned high and intermediate temperature oxidation conditions where low temperature reaction pathways could be ignored.

There have been numerous chemical kinetic mechanisms developed to describe chemical kinetics of $n\text{-C}_7\text{H}_{16}$ oxidation, a few of which are described below. Some early chemistry models were developed [22,23] to address temperatures above the NTC regime, and descriptions of low-temperature chemistry were added later. The mechanism developed by Curran et al. [24] included 25 reaction classes involving $n\text{-C}_7\text{H}_{16}$ as fuel and its immediate reaction products, as well as an extensive low-temperature kinetic submechanism. Côme et al. [22] used a computer package to generate detailed and semi-detailed high- and low-temperature chemical kinetic mechanisms for $n\text{-C}_7\text{H}_{16}$. More recent kinetic mechanisms, including $n\text{-C}_7\text{H}_{16}$ oxidation at high and low temperatures, have been developed [25,26] with considerable attention to alkylperoxy radical isomerization reactions that led to the recent study of Herbinet et al. [9]. A mechanism developed at Politecnico di Milano [27] includes low-temperature chemistry, building on a previous mechanism [23], and contains more than 13,000 reactions and 400 chemical species to describe the kinetics of alkanes with up to sixteen carbon atoms. In order to study gasoline surrogate mixtures, Mehl et al. [28] updated the Curran et al. [24] mechanism by incorporating improved kinetic rates for linear alkenes.

Minetti et al. [13] remains the only study to date with extensive sampling of the intermediates of $n\text{-C}_7\text{H}_{16}$ consumption in an RCM. Given the importance of $n\text{-C}_7\text{H}_{16}$ as a reference compound for the combustion chemistry of long-chain n -alkanes, the present work builds on these previous experimental and kinetic modeling studies by presenting new measurements and analysis of intermediate species formed during the ignition delay of $n\text{-C}_7\text{H}_{16}$ at high pressure and low temperatures. In the present study, an upgraded chemical kinetic model based on past work but incorporating the results of recent theoretical *ab initio* studies is used to simulate the new experimental data and test the predictive capabilities of the model.

Given the increasing interest in utilizing low-temperature heat release of fuels and bio-based fuels in advanced engine strategies such as homogeneous charge compression ignition, this project provides a new set of baseline experiments and kinetic modeling results for $n\text{-C}_7\text{H}_{16}$ that can be used to study combustion of mixtures of $n\text{-C}_7\text{H}_{16}$ and selected alcohol fuel species. The idea is to develop some kinetic understanding of fuel behavior in internal combustion engines when a biofuel (e.g., n -butanol) is blended with conventional petrochemical fuels such as gasoline, diesel fuel or jet fuel, for which $n\text{-C}_7\text{H}_{16}$ can be used as a surrogate component [29].

2. Experimental setup

2.1. The University of Michigan Rapid Compression Facility

As a unique and powerful apparatus for studying fuel chemistry over a broad range of thermodynamic conditions, the University of Michigan Rapid Compression Facility (UM RCF) has been used to study ignition of reference hydrocarbon fuel compounds such as *iso*-octane [30,31], simulated syngas mixtures of hydrogen and carbon monoxide [32], and oxygenated hydrocarbons [33], as well as soot formation and morphology [34,35]. The UM RCF has been applied to obtain time-resolved measurements of hydroxyl radical formation during *iso*-octane/air ignition [36], and quantitative measurements of the intermediate species of *iso*-octane [37], methyl butanoate [38], and n -butanol [39] oxidation using gas chromatography and rapid gas sampling.

The UM RCF is a chemical reactor that uses a free-piston/cylinder compression process to create the thermodynamic conditions necessary for combustion chemical kinetic studies, the features of which are interrogated using the optical and physical access provided by the test section of the facility. The five major components of the UM RCF, seen in the top panel of Fig. 1, are the driver section (with an inner diameter of 154 mm), the driven section (2.74 m long, 101.2 mm inner diameter), the test section (or test manifold), the sabot (a free piston with a tapered nosecone) and the fast-acting hydraulic globe valve (with a typical cycle time of 100 ms) system. Experiments begin with delivering a pre-prepared fuel/oxidizer/diluent mixture into the evacuated driven section, with the sabot located at the upstream end of the driven section. High-pressure gas in the driver section is released by the globe valve and launches the sabot down the length of the driven section, compressing the test gas mixture. At the end of compression (EOC), the nose cone of the sabot seals the fuel/oxidizer/diluent mixture in the test section, through an interference fit, at specifically targeted EOC temperatures and pressures. The thermodynamic conditions are achieved by varying the compression ratio of the test section, as well as the composition of gases in the test mixture. The majority of the temperature rise ($\sim 50\%$) and pressure rise ($\sim 80\%$) occur during the last 10 ms of the 145 ms compression stroke. After EOC, the fuel/oxidizer/diluent mixture in the test section autoignites after a period of time that is designated the ignition delay time (τ_{ign}), which is a function primarily of the test gas mixture composition, temperature and pressure.

The UM RCF is designed to create nominally uniform conditions at EOC to isolate reaction chemistry during the experiments and to minimize interfering effects. Two important features of the UM RCF—the geometry of the convergent section and the sabot—are critical to isolating reaction chemistry from the effects of fluid mechanics and heat transfer. Specifically, the unique shapes of the sabot and the convergent section allow the nearly isentropic compression of the gases in the center or core region of the driven section into the test section, while the cold boundary layer that is shed during the compression process is sealed from the core gases,

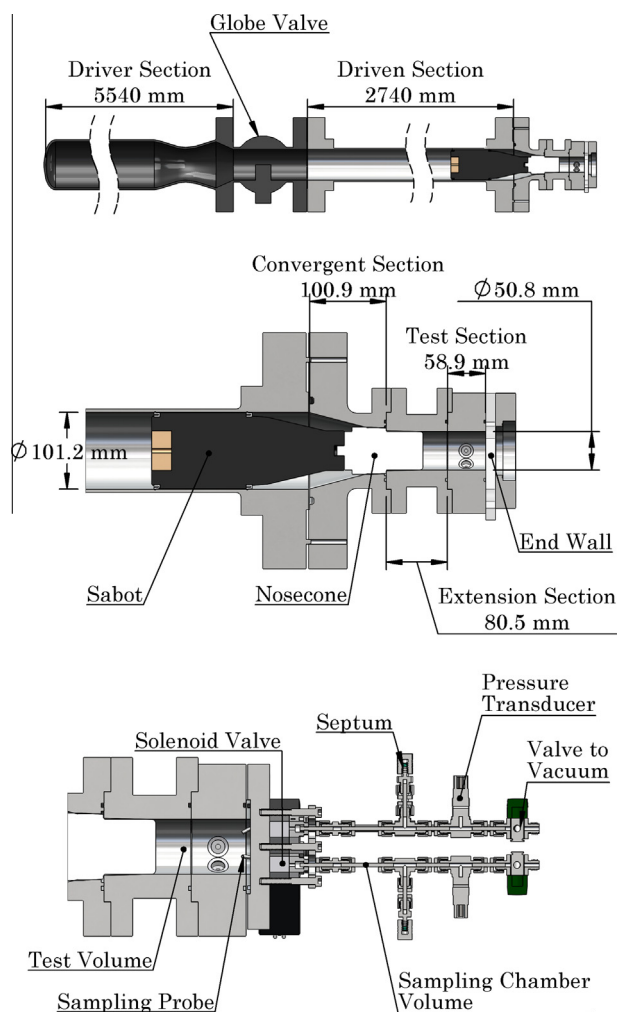


Fig. 1. Shown are a representation of the UM RCF with key dimensions (top panel), the configuration of the RCF test section for end-view imaging (middle panel), and the configuration of the RCF test section for high-speed gas sampling (bottom panel).

remaining trapped outside the test section in the space between the sabot and the convergent section. Heat losses from the gases in the core region are thus minimized, maximizing the amount of time the test gases are at uniform state conditions. Characterization studies of the UM RCF show the difference in measured and predicted isentropic conditions in the core region of the test section are less than 5% and the isentropic core region extends across 70% of the diameter of the test section [40]. These characteristics allow long test times (on the order of 50 ms, depending on test gas mixtures) during which pressure and temperature conditions remain at >75% and >80% of their EOC values, respectively [40].

For this study, stoichiometric $n\text{-C}_7\text{H}_{16}/\text{O}_2$ mixtures, with an inert/ O_2 diluent ratio of 5.64 (mole basis), were prepared manometrically in a magnetically-stirred mixing tank external to the UM RCF using a mixing manifold ($n\text{-C}_7\text{H}_{16}$ —Sigma—Aldrich, puriss. p.a., $\geq 99.5\%$, GC grade; O_2 —Cryogenic Gases, Purity Plus 4.3, 99.993%, <40 ppm Ar, <3 ppm moisture, <10 ppm N_2 , <0.5 ppm hydrocarbons; inert diluents: CO_2 —Cryogenic Gases, Laser Grade, 99.995%, <1 ppm of O_2 , moisture and hydrocarbons, <0.2 ppm CO, and N_2 —Cryogenic Gases, Purity Plus 5.0, 99.999%, <2 ppm O_2 , <3 ppm moisture, <0.5 ppm hydrocarbons). Mixture compositions were determined using partial pressures measured with a capacitance diaphragm gauge (Varian CeramiCel VCMT12TFA, with

an accuracy of ± 0.01 torr). The partial pressure of $n\text{-C}_7\text{H}_{16}$ was maintained well below its saturation vapor pressure at room temperature (0.059 atm or 45 torr at 25°C) in order to avoid concerns of fuel condensation. Total mixture pressures in the mixing tank were 0.5–0.8 atm, with initial fill pressures (P_0) in the RCF of approximately 0.13 atm.

The pressure in the test section is monitored using a piezoelectric transducer (Kistler 6041AX4) and charge amplifier (Kistler 5010B) with a combined accuracy of 0.01 atm and 0.015 ms. All electronic signals were recorded at 100 kHz (National Instruments cDAQ 9172 chassis coupled with National Instruments 9215 cards). For ignition experiments, a transparent end wall (configuration seen in the middle panel of Fig. 1) coupled with a high-speed camera with a widescreen CMOS array (Vision Research Phantom v711, maximum resolution of 1280×800 pixels, capable of 1.4 million frames per second at reduced resolution, $20\text{ }\mu\text{m}$ pixels with $0.34\text{ }\mu\text{m}$ spacing) and a fast 50 mm lens ($f/0.95$, Navitar) and c-mount extension tube recorded the ignition events in the test section by viewing along the axis of the test section. Imaging data provide qualitative and quantitative indications of the ignition homogeneity. Imaging data were acquired at 30,000 frames per second (fps) at a spatial resolution of 256×256 pixels, with an image exposure time of 34 μs .

2.2. High-speed gas sampling and gas chromatography

For gas sampling, the transparent end wall is replaced with an endwall equipped with a high-speed gas sampling system (shown in the bottom panel of Fig. 1). The gas sampling system removes a very small portion (quantified below) of the test gas mixture at targeted times during the ignition delay period. The samples are then analyzed using gas chromatographic techniques to identify and quantify the intermediate species present in sample. A series of samples acquired from experiments conducted at the same targeted EOC conditions allows a compilation of overall species time-histories for the intermediates identified. Several previous studies, including fuels such as iso-octane [37], methyl butanoate [38], and n -butanol [39] have demonstrated the validity of this experimental technique.

The sampling system used for these experiments consisted of sample chambers ($\sim 4.5 \pm 0.5$ mL) equipped with piezoresistive pressure transducers (Kistler 4045A2), charge amplifiers (Kistler 4618A0), low-bleed septa (VICI Valco), fast-acting sampling valves (modified Festo MHE3 valves with a stock response time of 3 ms, 3 mm orifice), and vacuum isolation valves (Swagelok). Independent control of each sampling valve using two digital delay/pulse generators (Stanford Research Systems Model DG535) allows the collection of up to four samples per experiment. The four samples are acquired through probes located on a square spacing (26×26 mm) on the end wall, 19 mm (radially) from the center of the test section. (Recall the test section inner diameter is 50.8 mm.) For the sampling data presented here, one gas sample was collected in each experiment by triggering the fast-acting valve located at the northeast position of the sampling end wall. Significant care was taken to ensure that the samples were taken from the volume of the test section well beyond the cold thermal boundary layer of the test section end wall; specifically, the sampling tips extend 1.2 cm into the core volume of the test section, whereas the boundary layer thickness is approximately 0.3 cm (see Fig. 2 described below). During the sampling event, the reacting gases removed from the test section are quenched in less than 0.3 ms as they expand into the vacuum of the sample chamber ($P_{\text{test section}}/P_{\text{sample chamber}} \sim 11$, $V_{\text{test section}}/V_{\text{sample chamber}} \sim 39$). Once the samples are acquired, the gases are drawn into a syringe (Hamilton Gastight #1010, 10 mL) through the syringe port on the sampling chamber for delivery to the gas chromatographs. As will

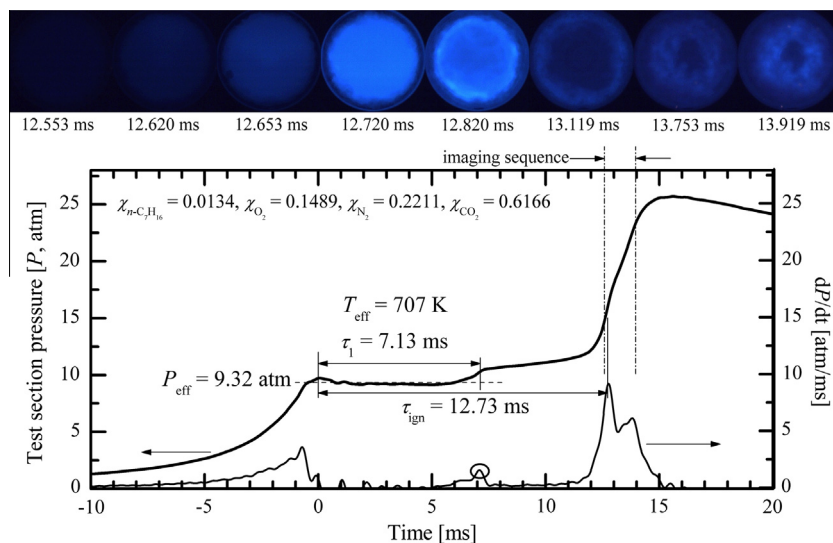


Fig. 2. Results from a typical UM RCF n -C₇H₁₆ ignition experiment. The lower panel depicts the pressure time-history in the test section, along with the rate of pressure rise, which allows definition of τ_{ign} and the effective pressure and temperature conditions. The upper panel shows still images taken at 30,000 fps via end-view imaging. Note the homogeneity of the ignition event. The color of the images has been adjusted for clarity. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

be shown later, the sampling of very small volumes of test gases leaves the reactive mixture in the test section unaffected. The τ_{ign} values determined from the gas-sampling experiments were in excellent agreement with experiments where gas sampling was not used. Further details on the gas sampling system can be found in Karwat et al. [39].

The dilution of the sample by unreacted gases trapped in the “dead” volume of the sampling system of the sampling probe ($\pm 16\%$, as determined previously [39]), along with the gas chromatography calibration uncertainties for each species, are the chief contributors to the uncertainty of the gas sampling measurements. For the data presented in this paper, the temporal uncertainties resulting from the triggering of the fast-acting sampling valves were the same as described previously [39]—the uncertainty in the sampling time is ± 0.75 ms centered on the falling edge of the sampling pulse sent to the fast-acting valves. The gas-sampling results therefore represent the average values of the species during the sampling time.

Three gas chromatographs (GCs) equipped with four different columns, with each connected to a separate detector (either a flame ionization detector, FID, or a thermal conductivity detector, TCD), were calibrated for quantitative measurements of species

of interest. A temperature-controlled 10-port gas sampling valve injected the samples into the columns in the GCs. Ultra high purity helium (Cryogenic Gases, Purity Plus, 99.999%) was the carrier gas for all of the GCs. Each of the FIDs used a hydrogen/air flame, were maintained at 300 °C, and were set with a range and attenuation of 1. The TCD was maintained at 100 °C with an attenuation of 1 and current of ± 160 mA. The helium, air, and hydrogen were further purified before use in the GCs using adsorbents to remove water, hydrocarbons, and oxygen. High-purity reference chemicals, either gaseous or vapors of liquid, were used to develop the GC temperature programs applied in the study and to calibrate the targeted intermediate species. (Details on purity levels of reference chemicals are provided in Table S1.) Calibration mixtures were made in the magnetically-stirred mixing tank. Upper limits to the calibrations for all species except n -C₇H₁₆ were established using the maximum concentrations predicted by the reaction mechanism discussed below during the ignition delay period of a mixture with $\chi(n\text{-C}_7\text{H}_{16}) = 0.0135$, $\chi(\text{O}_2) = 0.1486$, $\chi(\text{N}_2) = 0.2179$, and $\chi(\text{CO}_2) = 0.62$ at $P = 9$ atm and $T = 700$ K, and the maximum amount of carbon in the system. Calibration curves were linear in all cases. Voltage signals from the GC detectors were recorded using a data acquisition system with a sampling rate of 8 Hz (NI

Table 1
Experimental details of the gas chromatography systems used in the current work.

GC	System	Column(s)	Detector(s)	Temperature profile	Species detected
1	Perkin Elmer Autosystem	Varian CP-PoraBOND Q 25 m $\times$ 0.53 mm $\times$ 0.7 μ m	Flame ionization detector (FID)	30 °C (4.5 min) $\rightarrow$ 45 °C/min $\rightarrow$ 110 °C (9 min) $\rightarrow$ 45 °C/min $\rightarrow$ 150 °C (20 min)	n -C ₇ H ₁₆ , methanol (CH ₃ OH), acetaldehyde (CH ₃ CHO), propionaldehyde (C ₂ H ₅ CHO)
2	Perkin Elmer Autosystem	Varian CP-Al ₂ O ₃ /Na ₂ SO ₄ 25 m $\times$ 0.53 mm $\times$ 0.7 μ m	FID	30 °C for 4 min $\rightarrow$ 25 °C/min $\rightarrow$ 150 °C for 7 min $\rightarrow$ 45 °C/min $\rightarrow$ 200 °C for 2 min	Methane (CH ₄), ethane (C ₂ H ₆), ethene (C ₂ H ₄), ethyne (C ₂ H ₂), propane (C ₃ H ₈), propene (C ₃ H ₆), 1-butene (1-C ₄ H ₈), 1-pentene (1-C ₅ H ₁₀), 1-hexene (1-C ₆ H ₁₂), 1,3-butadiene (1,3-C ₄ H ₆), 3-heptene (3-C ₇ H ₁₄)
3	Perkin Elmer Clarus 500	Agilent DB-Wax 30 m $\times$ 0.25 mm $\times$ 0.25 μ m	FID	25 °C for 5 min $\rightarrow$ 45 °C/min $\rightarrow$ 200 °C for 5 min	2-Heptene (2-C ₇ H ₁₄), n -butyraldehyde (n -C ₄ H ₇ CHO)
		Restek ShinCarbon ST packed 2 m $\times$ 1 mm	Thermal conductivity detector (TCD)	25 °C for 5 min $\rightarrow$ 45 °C/min $\rightarrow$ 200 °C for 5 min	Carbon monoxide (CO)

PXI 4472). Table 1 provides details about the temperature profiles and columns used in the GCs and the species detected.

3. Experimental results and discussion

3.1. Ignition experiments

Experiments with end-view imaging were performed not only to measure the ignition delay time (τ_{ign}) as a function of temperature, but also to identify the appropriate temperature and pressure conditions at which to conduct sampling experiments. Given that each sampling event lasts approximately 1.5 ms, we targeted a τ_{ign} of approximately 15 ms for the sampling experiments. This time would comfortably allow for at least 10 discrete sampling events during the ignition delay period. The dilution in these experiments (inert/ $\text{O}_2 = 5.64$ rather than the more commonly used ratio of 3.76 in normal air) was selected to extend the ignition delay period to 15 ms, as discussed further below. For each experiment, the effective test conditions corresponding to a τ_{ign} value are determined based on the pressure time-history, an example of which is seen in Fig. 2. $n\text{-C}_7\text{H}_{16}$ exhibited two-stage ignition for all of the experimental data presented here. Therefore, a modified method compared to previous studies [38,39] was used to describe the experimental conditions. Eq. (1) was used to determine the effective pressure (P_{eff}), which is the time-integrated average pressure from the maximum pressure (P_{max}) at the EOC to the maximum rate of pressure rise at the 1st stage of ignition ($dP/dt_{\text{max, 1st stage}}$),

$$P_{\text{eff}} = \frac{1}{t_{dP/dt_{\text{max}}} - t_{P_{\text{max}}}} \int_{t_{P_{\text{max}}}}^{t_{dP/dt_{\text{max, 1st stage}}}} P \cdot dt \quad (1)$$

The effective temperature (T_{eff}) for each experiment was determined using P_{eff} and numerical integration of the isentropic relation (Eq. (2)),

$$\int_{T_0}^{T_{\text{eff}}} \frac{\gamma}{\gamma - 1} d \ln T = \ln \left(\frac{P_{\text{eff}}}{P_0} \right) \quad (2)$$

where P_0 is the initial charge pressure, T_0 is the initial temperature, and γ is the temperature-dependent ratio of the specific heats of the unreacted test gas mixture (determined using the NASA thermodynamic data base [41]). τ_{ign} for each experiment was defined as the time between EOC ($t = 0$ ms, defined by the first maximum in P) and the maximum rate of pressure rise corresponding to autoignition ($dP/dt_{\text{max, 2nd stage}}$). Ignition experiments were performed in the narrow temperature range of 660–707 K at an effective pressure between 8.92 and 9.53 atm. The lowest EOC temperature was dictated by the compression ratio of the UM RCF in the configuration used, with CO_2 as the sole diluent. The highest EOC temperature was chosen to avoid significant reaction and heat release during compression, thereby avoiding complications in defining the experimental state conditions and the interpretation of the resulting τ_{ign} and gas sampling results. All mixtures used an equivalence ratio of 1 (based on the molar ratio of fuel to O_2) and a dilution of inert/ $\text{O}_2 = 5.62\text{--}5.64$, with N_2 and CO_2 as the diluents. The $n\text{-C}_7\text{H}_{16}$ concentration for the experiments was 1.34–1.35%. Table S2 of the Supplemental Information provides a summary of the experimental conditions and results for τ_{ign} for all of the UM RCF data presented.

Figure 2 presents typical results from a UM RCF $n\text{-C}_7\text{H}_{16}$ ignition experiment in which imaging data were acquired. The lower panel depicts the time-histories of the pressure (P) and rate of pressure rise (dP/dt) in the test section. A fast Fourier transform has been applied to filter high-frequency disturbances greater than 2.5 kHz generated by the impact of the sabot near EOC. A smooth compression process due to the motion of the sabot brings the pressure to the first maximum, and the EOC is set as time $t = 0$, after which the volume in the test section is constant. The first stage of ignition (τ_1) occurs at 7.13 ms, corresponding to a local maximum in dP/dt (circled at the bottom in Fig. 2), and P_{eff} and T_{eff} are 9.32 atm and 707 K, respectively. After the first stage of ignition, the pressure

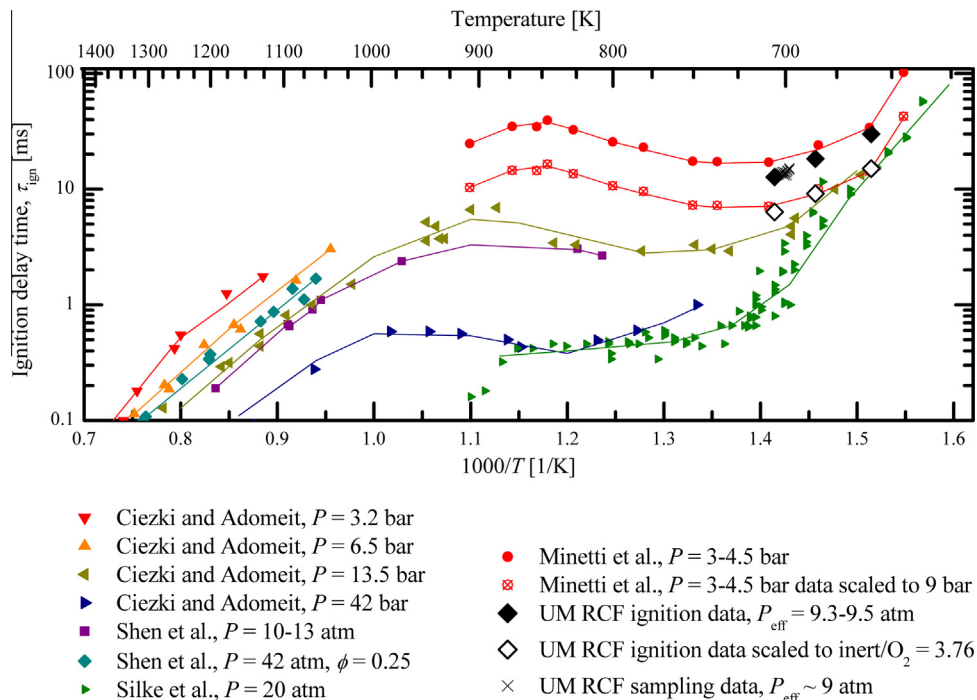


Fig. 3. Ignition characteristics of $n\text{-C}_7\text{H}_{16}$ over a wide range of temperatures (650–1400 K) and pressures (3–42 atm), including results of the current work. All data with $\phi = 1$ unless otherwise noted. UM RCF data with inert/ $\text{O}_2 \sim 5.64$, and all other data with inert/ $\text{O}_2 \sim 3.76$. Shock tube ignition delay data are represented at T_5 corresponding to P_5 in shock tubes, the Minetti et al. [13] RCM ignition data are represented at T_{EOC} , and the data of the current work are represented at T_{eff} corresponding to P_{eff} . Error bars representing the standard deviation of τ_{ign} (0.63 ms) for the UM RCF data are not visible at the resolution used in the plot. Lines represent curve fits to experimental data.

risers again slowly for a while, then more abruptly, to the maximum value corresponding to the autoignition of the test mixture— $\tau_{\text{ign}} = 12.73$ ms.

The upper panel of Fig. 2 shows stills from the image sequence of the chemiluminescence that occurs during ignition. This emission is attributed [42] to CH^* ($\text{C}_2 + \text{OH} \rightarrow \text{CO} + \text{CH}^*$) and C_2^* ($\text{C} + \text{CH} \rightarrow \text{C}_2^* + \text{H}$) radicals, which have strong spectroscopic features in the blue part of the visible spectrum (CH^* : 431.2 nm; C_2^* : 473.7 nm, 516.5 nm, 563.5 nm). The intense blue emission occurs simultaneously throughout the test section with uniform intensity, attesting to the high degree of homogeneity of the reactant mixture and the state conditions in the test section. Such uniformity gives confidence in the application of localized sampling.

Figure 3 presents results of UM RCF ignition and sampling experiments, along with $n\text{-C}_7\text{H}_{16}$ ignition data from other studies, including RCM studies of Minetti et al. [13] and Silke et al. [10], and shock tube studies of Shen et al. [2] and Ciezki and Adomeit [1]. Data for this plot have been selected to illustrate the influences of pressure, equivalence ratio, and dilution on τ_{ign} and to show that the present RCF experimental results are consistent with previous ignition delay experiments using $n\text{-C}_7\text{H}_{16}$ as the fuel. As shown in Fig. 3, $n\text{-C}_7\text{H}_{16}$ ignition exhibits significant NTC behavior between 720 K and 950 K, and the NTC region shifts to higher temperatures as pressure increases. Reasons for this shift are well known; NTC behavior is governed by the ratio of RO_2 to R , and since at higher pressures the concentrations of O_2 are higher, the equilibrium is shifted towards RO_2 formation, consequently moving the NTC region to higher temperatures [43]. The magnitude of the NTC behavior also decreases with increasing pressure, illustrated in Fig. 3 by the Ciezki and Adomeit results at 42 and 13.5 bar pressure [1]. The decrease in NTC behavior with increasing pressure, discussed by Ciezki and Adomeit [1] and by Westbrook et al. [57], can be best attributed to the quadratic increase with pressure of the bimolecular reaction rates, which release most of the energy in the autoignition. This is in contrast to the equilibrium of the $\text{R} + \text{O}_2 = \text{RO}_2$ and $\text{QOOH} + \text{O}_2 = \text{O}_2\text{QOOH}$ reactions, reactions that control the NTC behavior and are equilibrated in most of the NTC region, which shift only linearly with increasing pressure.

Also seen in Fig. 3 are data from the ignition delay experiments of Shen et al. [2], which show that at temperatures between approximately 1050 and 1250 K, fuel-lean mixtures take longer to ignite than the stoichiometric mixtures. The authors reported similar behavior for $n\text{-decane}$, $n\text{-dodecane}$ and $n\text{-tetradecane}$ in addition to $n\text{-C}_7\text{H}_{16}$, with lean mixtures igniting more slowly than stoichiometric mixtures, an effect consistent for many hydrocarbon fuels at lower temperatures [1,30,44–46]. The results of Minetti et al. [13] are shown as two curves, the upper one in Fig. 3 showing the results of the experiments actually carried out at pressures from 3 to 4.5 bar, and the lower curve showing the same results scaled to a pressure of 9 bar (using a scaling proportional to P^{-1} following Davidson et al. [58]) in order to compare them with the present experiments, which were conducted at an effective pressure close to 9 bar.

The values measured for τ_{ign} in the current study are shown as the three filled black diamonds in Fig. 3. The rather limited temperature range of these experiments is evident from the figure, but the trend seen in the slope of the three experimental points clearly matches the slope of the Minetti et al. [13] and Ciezki and Adomeit [1] results over the same temperature range. However, the present experimental ignition delay times, measured at an effective pressure of approximately 9 bar, are somewhat longer than the shock tube values of Ciezki and Adomeit [1] at the comparable pressure of 13.5 bar and similarly longer than the RCM results of Minetti et al. [13] when scaled to 9 bar. The differences are attributed to the higher dilution used in the present work. As noted above, the UM RCF data were obtained using an inert/ O_2 molar dilution of

~5.64, while the other studies summarized in Fig. 3, as well as the experiments of Griffiths et al. [12], were obtained using an inert/ O_2 molar dilution of 3.76, i.e. the N_2/O_2 ratio in normal air. We used the current kinetic mechanism to compare computed ignition delay times for the same reactive mixtures with dilutions of 3.76 and 5.64, indicating a difference of approximately a factor of 0.5, concluding that the higher diluent levels in our experiments produced ignition delay times twice as long as would be observed if the amount of diluent was 3.76, the value of the other experimental studies reported in Fig. 3. Scaling the three UM RCF experimental points in Fig. 3 by this ratio resulted in the open diamond symbols shown in Fig. 3, which are in excellent agreement with the other experiments at comparable conditions.

3.2. High-speed gas sampling experiments

While ignition studies provide a good test of the global kinetics of $n\text{-C}_7\text{H}_{16}$ ignition, speciation measurements require a more detailed understanding of the important chemical pathways in the reacting test gas mixture and place stricter constraints on the chemical kinetic reaction mechanism used to simulate the experiments. Intermediate species formed during the ignition delay time were measured through gas sampling experiments. Figure 4 shows results from a typical sampling experiment, in which $P_{\text{eff}} = 9.01$ atm, $T_{\text{eff}} = 700$ K, $\tau_1 = 8.49$ ms and $\tau_{\text{ign}} = 14.53$ ms, showing the pressure time-history in the test section, the electronic pulse used to trigger the high-speed gas sampling system, and the pressure in the sampling chamber. Since only a very small amount of sample is removed from the reacting mixture in the test section, the pressure in the test section remains unaffected by the sampling process. The pressure features seen in Fig. 4 are thus very similar to those seen in Fig. 2, where no sampling was employed.

Also shown as the dashed curve in Fig. 4 is the time-history of a non-igniting experiment, in which the O_2 of an igniting experiment was replaced with N_2 . Given the almost identical thermal characteristics of O_2 and N_2 , the mixture compressed in a non-igniting experiment provides a baseline for comparison, including an understanding of the effects of heat transfer during the experiments. As seen in Fig. 4, the pressure time-history of the non-igniting experiment is almost indistinguishable from the igniting experiment (P_{eff} and T_{eff} differ by less 0.1% and 0.3%, respectively) until the first stage of ignition, demonstrating that the heat transfer physics of the experiments remain unaffected by the sampling events. Moreover, because the non-igniting experiment shows a nearly identical compression process to the igniting experiment,

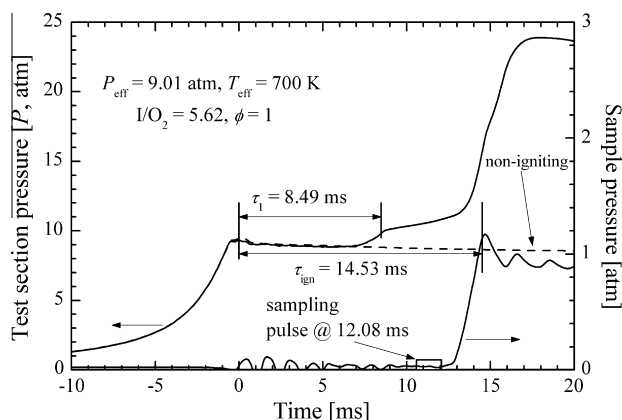


Fig. 4. Typical pressure time-histories from the test section and sampling volume of a gas-sampling experiment. The electronic signal used to trigger the gas sampling event is included for reference. Also included in the figure is the pressure time-history of a non-igniting experiment. See text for details.

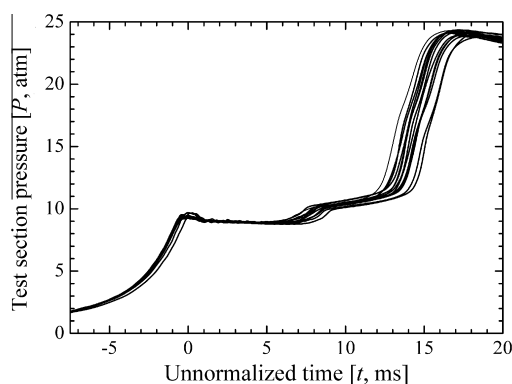


Fig. 5. The experimental pressure time-histories of the thirteen gas-sampling experiments. Although unnormalized, note the level of repeatability of the compression process, as well as the first and second stages of ignition and heat release.

the data support that reaction during compression is not a concern in these experiments.

A summary of the gas-sampling experiments, including mixture compositions, can be found in Table S2 in the Supporting Information. Figure 5 shows the pressure time-histories for the thirteen experiments, all of which show nearly identical, smooth compression processes, and very similar pressure time-histories after EOC. The average P_{eff} , T_{eff} , τ_1 , and τ_{ign} for all of the gas-sampling experiments were 9.02 atm, 701 K, 7.94 ms, and 14.09 ms, respectively, with corresponding standard deviations (representing the uncertainty of the data) of 0.07 atm, 1.5 K, 0.52 ms, and 0.63 ms, respectively, which demonstrate the excellent repeatability of the experiments. Furthermore, since gas-sampling and end-view imaging cannot be conducted simultaneously, an experiment to confirm spatial homogeneity of the reacting mixture in the test section was performed by rotating the gas sampling system by 180° and acquiring a sample from the southwest corner of the test section at almost the same time during the ignition delay period as a sample acquired in another experiment from the northeast corner. The data from the rotated gas sampling system were within measurement uncertainties of the baseline (i.e. unrotated) species concentrations.

For comparison with kinetic mechanism predictions, the pressure time-histories and sampling times were converted to normalized times. The period of time between EOC and first stage of ignition of each experiment was normalized by τ_1 (resulting in a normalized time domain of 0 to 1), and the period of time between

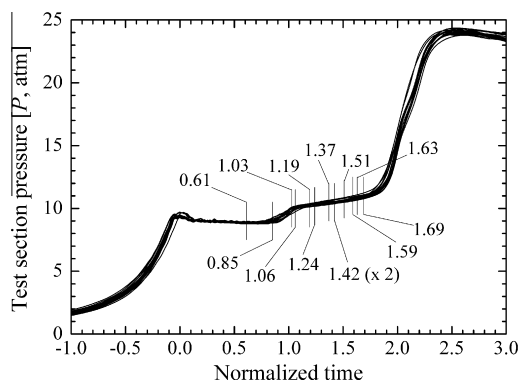


Fig. 6. The normalized experimental pressure time-histories of the thirteen gas-sampling experiments. 0 represents EOC, 1 represents the first stage of ignition, and 2 represents the second stage of ignition. Shown also are the normalized sampling times of each experiment.

the first stage of ignition and autoignition ($\tau_{\text{ign}} - \tau_1$) was normalized by $\tau_{\text{ign}} - \tau_1$ and added to the first normalized time domain. The result, seen in Fig. 6, is an overall normalized time domain in which 0 to 1 represents the first stage of ignition, and 1 to 2 represents the second stage of ignition.

Figure 7 presents typical chromatograms obtained from the GCs during a sampling experiment, specifically, the experiment depicted in Fig. 4. All identified species except CO are shown in Fig. 7. C_2H_2 was not observed in any experiment, and some peaks remained unidentified. The carbon balance was $90 \pm 12\%$ for sampling at early times during the ignition delay period and $65 \pm 9\%$ for sampling closer to autoignition. The peaks identified in the chromatograms were converted into discrete measurements of intermediate species for each normalized sampling time using the calibrations for each species.

Several peaks corresponding to large hydrocarbons and oxygenated compounds remained unidentified in our GC-FID analyses. In particular, we were unable to identify from the GC-FID analysis any large, C_7 oxygenated species such as furans, cyclic ethers, or ketohydroperoxides (chemical standards were difficult to find for many of these species). The Minetti et al. study [13] detailed results of classes of compounds such as furans and oxirane, as well as large ketones. The kinetic model, described below, showed appreciable levels of these species during the later stages of the ignition delay period. For example, just before the onset of the first stage ignition, when the computed $n\text{-C}_7\text{H}_{16}$ mole fraction had decreased from its initial value of 0.0135 to 0.0059, the model showed a sum of the C_7 cyclic ether mole fractions of about 0.002 and a total of about 0.0005 of C_7 ketohydroperoxides, equal to about 32% of the $n\text{-C}_7\text{H}_{16}$ consumed at this point. Based on these results, it is likely that much of the carbon missing from the experimental carbon balance is in the form of these large C_7 oxygenated species, whose concentrations increase steadily during the ignition delay period.

Figure 8 presents as symbols (black circles) the UM RCF species measurements during the ignition delay time of $n\text{-C}_7\text{H}_{16}$ for the average experimental conditions of $P_{\text{eff}} = 9.02$ atm, $T_{\text{eff}} = 701$ K, $\chi(n\text{-C}_7\text{H}_{16}) = 0.0134$, $\chi(\text{O}_2) = 0.149$, $\chi(\text{N}_2) = 0.2336$, and $\chi(\text{CO}_2) = 0.6040$, where the average ignition features occur at $\tau_1 = 7.94$ ms and $\tau_{\text{ign}} = 14.09$ ms. The ± 0.75 ms uncertainty in the sampling times, when normalized by the average τ_{ign} , corresponds to approximately ± 0.053 . The uncertainty in the species calibrations (varying for each species) and the uncertainty due to pressure measurements in the sampling volume ($\pm 10\%$) were considered as independent sources of uncertainty on the species concentrations, and were therefore combined by using the square root of the sum of the squares. The quenching and recombination of small radicals can also be considered a source of error or uncertainty in the experimental measurements. Small radicals such as O, H, OH, HO_2 , and CH_3 can recombine to form water vapor and small hydrocarbons, which may interfere with measurements of stable species. However, predicted small radical mole fractions, seen in Fig. 9, are very low ($\sim 40\text{--}100$ ppm) relative to most of the measured intermediate species before autoignition. Radical recombination during the quenching process of sampling is thus not expected to be a source of error in the species measurements presented in this work.

Seen also in Fig. 8 (open red squares) are measurements of the species time-histories reported by Minetti et al. [13] in their ignition and speciation study. Their experiments were conducted with stoichiometric mixtures of $n\text{-C}_7\text{H}_{16}$ and O_2 with an EOC pressure and temperature of 3.4 bar and 667 K, resulting in an average τ_{ign} of 41 ms, with τ_1 around 30 ms. The sampling experiments by Minetti et al. [13] were at a similar temperature to the present UM RCF experiments, but at a lower pressure. At 3.76, the dilution in their experiments was lower than the 5.64 dilution in the present study. Also, in contrast to the high-speed sampling used in the

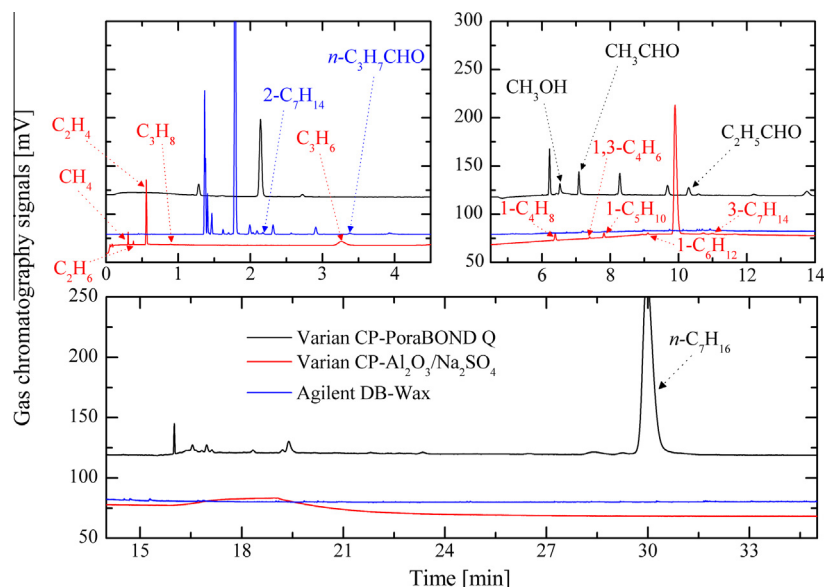


Fig. 7. An example of typical gas chromatograms of a gas sample from an $n\text{-C}_7\text{H}_{16}$ sampling experiment. The data are from the experimental results presented in Fig. 4, in which sampling was performed at 12.08 ms after EOC.

present study, Minetti et al. [13] quenched the entire contents of the combustion chamber in their experiments by bursting a diaphragm, thereby combining their measured species with the contents of a cold boundary layer. Nevertheless, the species results from the present study are broadly consistent with those of Minetti et al. [13].

4. Kinetic modeling

Chemical kinetic modeling is frequently used to interpret RCM experimental results, usually considering zero-dimensional constant volume, adiabatic model predictions using a detailed reaction mechanism for the fuel studied. For very reactive fuels, chemical reactions and heat release during the compression stroke can be included in the modeling calculations, and heat losses to combustion chamber walls can also be included, often by making small adiabatic expansions of the combustion volume after the end of the compression stroke to simulate the cooling of the reacting gases. In the present simulations, reactions during the compression stroke and simulated heat losses were examined, and neither process was found to have significant impacts on computed results. For example, taking the experimental pressure time-history as an indicator of heat loss, the pressure decreases less than 3% between EOC and the first stage of ignition, corresponding to a temperature decrease of less than 1%. This low heat loss is primarily due to the very high volume-to-surface area ratio of the test section. As a consequence, all simulations used a constant volume, adiabatic approach, and for comparison with the species measurements, all the simulations used a reactant mixture of $\chi(n\text{-C}_7\text{H}_{16}) = 0.0134$, $\chi(\text{O}_2) = 0.149$, $\chi(\text{N}_2) = 0.2334$, and $\chi(\text{CO}_2) = 0.6042$, and initial conditions of $P_0 = 9$ atm and $T_0 = 700$ K.

The kinetic modeling calculations were carried out using the CHEMKIN Release 10101 (x64) [47]. The core $\text{H}_2/\text{O}_2/\text{C}_1\text{--C}_4$ species mechanism was taken from a recently refined mechanism of Metcalfe et al. [48]. The kinetic reaction mechanism for $n\text{-C}_7\text{H}_{16}$ first selected for this study was developed by Mehl et al. [28], which was based on an earlier mechanism of Curran et al. [24]. Both mechanisms are constructed using 25 reaction rate rules for the distinct classes of reactions that describe the fuel oxidation. This approach, combining a core small molecule kinetic mechanism

with additional submechanisms for the larger fuels of interest in a specific study, is commonly employed by a majority of kinetic model builders. The result in the present case was tested by simulations of most of the validation targets used to first develop the earlier version of the reaction mechanism [24,28], and the present mechanism reproduced all of those tests in satisfactory fashion. Three examples of such reaction classes include the abstraction of H atoms from the fuel by small radical species such as OH or H,

$$n\text{-C}_7\text{H}_{16} + \text{X} = \text{C}_7\text{H}_{15} + \text{XH} \quad (\text{C-2})$$

where X represents the radical; the addition of molecular oxygen to an alkyl radical, a heptyl radical in the present case, to produce an alkylperoxy radical RO_2 ,



and isomerization reactions of RO_2 to produce a hydroperoxy alkyl species QOOH ,



These reactions are labeled according to the reaction classes defined by Curran et al. [24]. The complete kinetic mechanism contains approximately 1800 chemical species and 7250 elementary chemical reactions, and is included in the supplemental information.

Since most of the reactions in such a large kinetic mechanism have never been studied individually, their rates are usually estimated on the basis of other reactions that are somehow similar to the one being examined. For example, for reactions in class (C-10) above (there are 4 unique primary and secondary heptyl radicals in the present mechanism for $n\text{-C}_7\text{H}_{16}$), the rates are assumed to be equal to the rates of addition of O_2 to primary and secondary sites in n -butane or propane, reactions whose rates have been studied. However, for some reaction classes, there were no reliable reaction rates for any similar fuels at the time the original kinetic mechanisms were developed. One such class involves RO_2 isomerization reactions of class (C-12) above. Models for this class in the $n\text{-C}_7\text{H}_{16}$ mechanism of Curran et al. [24] were taken from alkyl radical isomerizations with the same size transition state rings by which the reaction proceeded and the same type of C–H bond being broken (i.e., primary, secondary or tertiary), but did not account for the –O–O– structure that is part of the transition

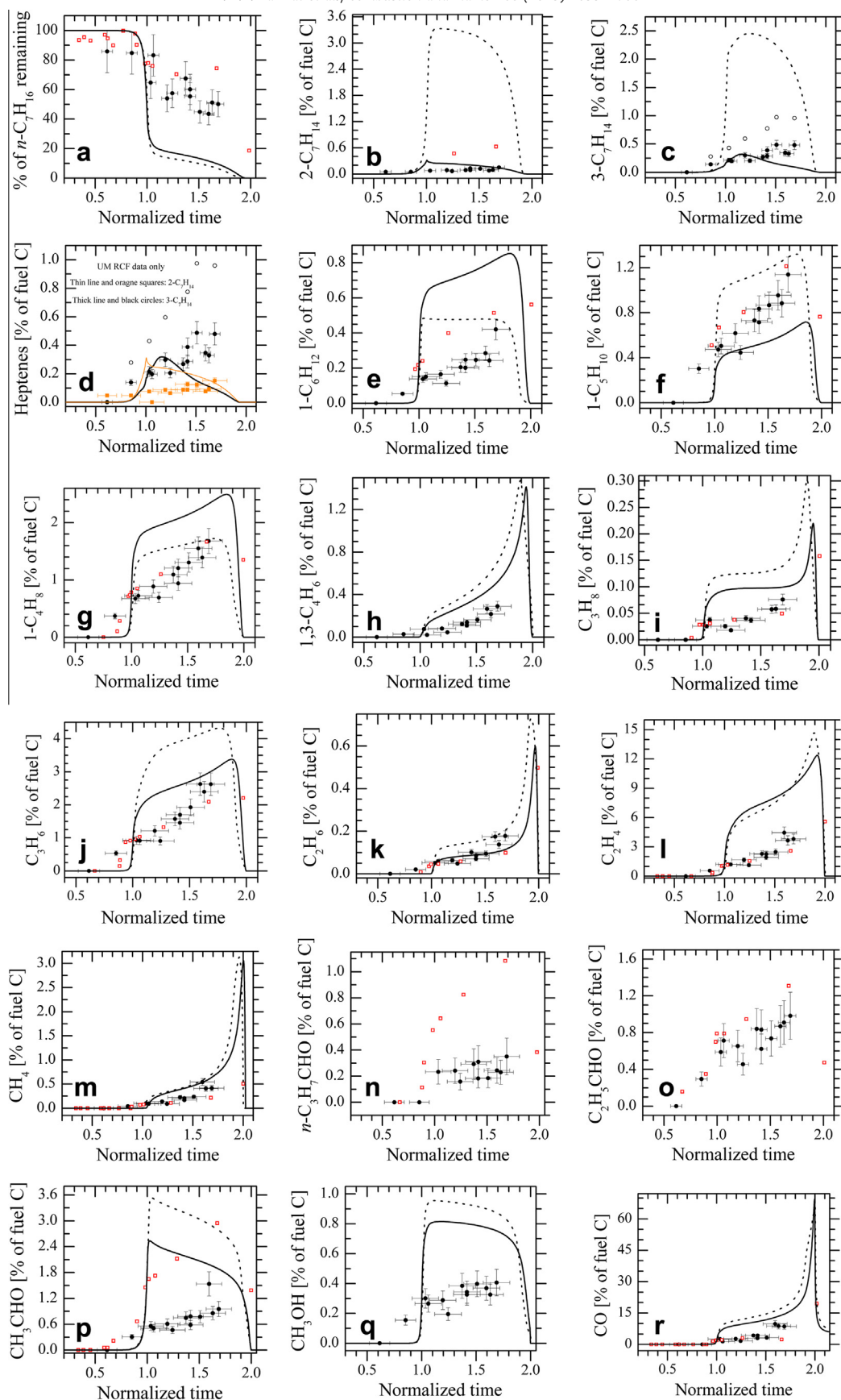


Fig. 8. (a–r) Species concentration time-histories from the UM RCF sampling experiments (solid black circles), from the Minetti et al. [13] study (open red squares), and from the model predictions for the UM RCF data based on the Mehl et al. [28] mechanism (dashed line), and the current, modified reaction mechanism (solid line). (Note the unique symbols for figure (d).) The model predictions are shown for constant volume, adiabatic conditions with $P_0 = 9$ atm, $T_0 = 700$ K, $\chi(n\text{-C}_7\text{H}_{16}) = 0.0134$, $\chi(\text{O}_2) = 0.149$, $\chi(\text{N}_2) = 0.2336$, and $\chi(\text{CO}_2) = 0.6040$. All data are presented as normalized to the % of fuel carbon in the initial reactant mixture. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

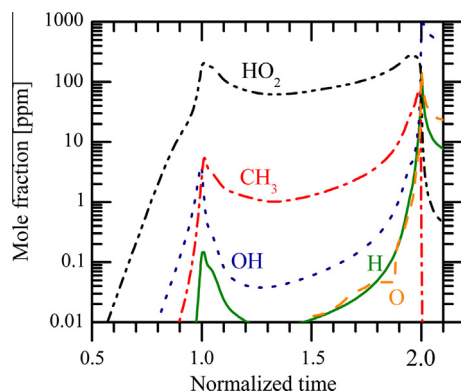


Fig. 9. Predictions (constant volume, adiabatic simulations with $P_0 = 9$ atm, $T_0 = 700$ K, $\chi(n\text{-C}_7\text{H}_{16}) = 0.0134$, $\chi(\text{O}_2) = 0.149$, $\chi(\text{N}_2) = 0.2336$, and $\chi(\text{CO}_2) = 0.6040$) using the current mechanism of radical concentrations at the first and second stages of ignition.

state ring in reaction class (C-12), which should be expected to influence the amount of ring strain energy in the transition state. This logic had been proposed by Pollard [49], based in turn on analysis from Fish [50] and was necessary because no rate data were available for any of the RO_2 isomerizations. The same type of approximation strategy was necessary for other reaction classes involving large hydrocarbon radicals that contained O atoms, including cyclic ether production from the hydroperoxy alkyl radical QOOH, class (C-19); QOOH decomposition to produce an alkene and HO_2 , class (C-20); QOOH β -decomposition to produce an alkene and a carbonyl species as well as an OH radical, class (C-21); addition of molecular oxygen to the QOOH species, class (C-22); and isomerization of the O_2QOOH radical to produce a stable ketohydroperoxide species and OH, class (C-23). The common feature of these reaction classes is that they are all part of the RO_2 isomerization sequence of reactions where low temperature (650–850 K) NTC behavior occurs.

Kinetic reaction mechanisms built according to past reaction rate rules, with many of their reaction rates estimated because of the absence of supporting experimental or theoretical values, have been quite successful in reproducing integrated experimental quantities such as ignition delay times and laminar burning velocities. However, in most such experiments, few if any intermediate species concentrations are reported that can provide much more challenging tests of the predictive capabilities of reaction mechanisms, especially in the low temperature regime where so many of the reaction rates are poorly known. As noted above, the

intermediate species measurements of Dagaut et al. [8] and Minetti et al. [13] from their jet-stirred reactor and RCM experiments, respectively, were used by Curran et al. [24] to build the original $n\text{-C}_7\text{H}_{16}$ kinetic mechanism, but those were the only previously existing such measurements that were suitable for mechanism validation. The recent jet-stirred reactor experiments of Herbinet et al. [9], together with the current RCF data, provide much-needed additional validation data for $n\text{-C}_7\text{H}_{16}$ chemical kinetic models, as these experimental data focus especially on intermediate chemical species concentration measurements in the low-temperature regime.

In this study, the mechanism developed by Mehl et al. [28] was used for initial calculations. Figure 10 presents the comparison of the initial model results and experimental results for pressure time history. The preliminary calculations produced two-stage ignition for the current experimental conditions, with first-stage and second-stage ignition delay times that were somewhat shorter than but within a factor of two of the experimental results, as shown in Fig. 10. Specifically, for the initial calculations at 700 K, the predicted τ_1 was 5.5 ms and τ_{ign} was 8.5 ms, compared with the experimental values of about 8.5 ms and 14.5 ms, respectively.

Results for the initial model predictions for the intermediate species are compared with the experimental measurements in Fig. 8 as the dashed lines. The initial calculations showed that approximately 80% of the $n\text{-C}_7\text{H}_{16}$ fuel was consumed in the first ignition stage, more than the approximately 40–60% consumption observed in the experiments, as seen in Fig. 8a. More dramatically, the computed results showed that the intermediate concentrations of the heptene intermediates produced from $n\text{-C}_7\text{H}_{16}$ during the time interval between the first and second ignition stages were higher than the experimentally measured levels by a factor of 5 to 25, as seen in Fig. 8b and c. The 3- C_7H_{14} calibration standard was an uncertain mixture of *cis*-3- C_7H_{14} and *trans*-3- C_7H_{14} and the measured areas of these two isomers were approximately equal in the chromatograms. Due to the uncertainty in the calibration standard, the 3- C_7H_{14} was quantified within a factor of two. In Fig. 8c and d, the solid black circles represent measurements of a single 3- C_7H_{14} isomer, and the open circles represent the upper bound/cumulative measurement of the 3- C_7H_{14} isomers.

The initial mechanism predicted many species quite accurately. For example, as seen in Fig. 8, CH_4 and C_2H_6 were well reproduced, as were CO and some small alkenes such as C_3H_6 and 1- C_4H_8 . Other key species such 1- C_5H_{10} , 1- C_6H_{12} and C_2H_4 , together with smaller oxygenates such as CH_3CHO and CH_3OH , were captured within factors of about two by the initial mechanism. However, the large errors in the computed amounts of the heptene isomers, together with the considerable differences between modeled and experimental rates

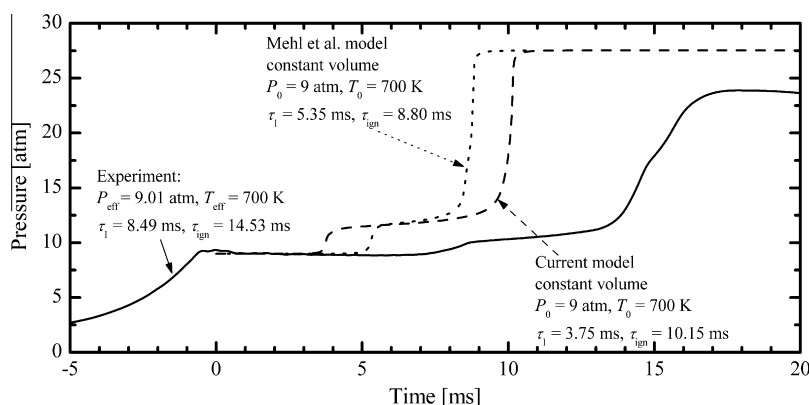


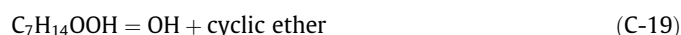
Fig. 10. A comparison of experimental and computational pressure time-histories. The models include predictions (constant volume, adiabatic simulations with $P_0 = 9$ atm, $T_0 = 700$ K, $\chi(n\text{-C}_7\text{H}_{16}) = 0.0134$, $\chi(\text{O}_2) = 0.149$, $\chi(\text{N}_2) = 0.2336$, and $\chi(\text{CO}_2) = 0.6040$) based on the current reaction mechanism and the Mehl et al. [28] mechanism.

of n -C₇H₁₆ consumption in the first stage ignition, indicated that improvements in the kinetic modeling of the low temperature portion of the reaction mechanism were needed.

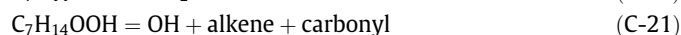
Two recent publications by Villano et al. [43,51] provided a basis for making significant improvements to the n -C₇H₁₆ kinetic mechanism. In the papers by Villano et al., the reaction rates for the most important reaction classes that control ignition at low temperature conditions were calculated with electronic structure calculations performed at the CBS-QB3 level of theory combined with canonical transition state theory calculations. The first of these papers [43] reported rates of addition of O₂ to large alkyl radicals to produce RO₂ (class (C-10) above); the dissociation of the RO₂ back to alkyl radicals and O₂ (the reverse reactions for class (C-10)); the isomerization of RO₂ to produce hydroxyalkyl radicals QOOH (class (C-12) above); and the concerted elimination reactions of RO₂ to produce alkenes and HO₂ radicals, seen below:



The second paper [51] extended the same approach to include the reverse reaction rates for hydroperoxyalkyl radicals QOOH back to RO₂ (the reverse direction for class (C-12) above); the cyclization of the QOOH species to produce a cyclic ether and OH



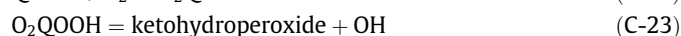
and β -scission reactions of QOOH to produce stable species and a radical:



All of these reaction rates were modified in the n -C₇H₁₆ kinetic mechanism, replacing the rates that had been estimated based on the discussions of Pollard [49].

In most kinetic reaction mechanisms and in the present work, the reaction rates in the mechanism are given in one direction and the rates of the reverse reactions are determined from the forward rates and detailed balance from the relevant thermochemistry. The reverse rates for some multistep reactions that are not genuine elementary reactions, such as Reaction (C-21) above, are usually set to zero, since the net reverse reaction cannot occur. The theory work of Villano et al. calculated the rates of elementary reactions in the absence of their corresponding reverse reactions, and some of their theoretical computations addressed both forward and reverse rates of truly elementary reactions. In particular, their work provides elementary rates of both Reaction (C-12) and its reverse. We chose to specify the forward and reverse reaction rates as determined by Villano et al.; in principle, this may not satisfy detailed balance and may not provide the correct equilibrium constants for these reactions. However, we felt that the precision of the forward and reverse reaction rate determinations was likely to be as accurate or better than our knowledge of the thermochemistry of the species involved. Further analysis of these theoretically determined reaction rates may lead to future improvements in the thermochemistry of the RO₂ and QOOH species.

However, several important reaction pathways in the low temperature submechanism were not evaluated by Villano et al., including reaction classes (C-22) and (C-23); the addition of molecular O₂ to the hydroperoxy alkyl radicals QOOH; and the isomerization reactions of the resulting O₂QOOH species to produce a relatively stable ketohydroperoxide species.



These last two reaction classes are especially important since chain branching at low temperatures cannot occur without the

second addition of O₂ to QOOH and the subsequent isomerization and decomposition reactions that produce multiple small radical species [24,52,53].

Fortunately, the work of Villano et al. [43,51] enables us to estimate the rates of reactions in classes (C-22) and (C-23) with much greater confidence than based on Pollard's estimates, as follows:

1. Since Villano et al. recommended rates of O₂ addition to primary, secondary and tertiary sites in alkyl radicals, we assumed that addition rates of O₂ to primary, secondary and tertiary sites in QOOH radicals would be approximately the same at each type of site as those for alkyl radicals.
2. Villano et al. calculated rates of RO₂ isomerization reactions by specifying the number of C, O, and H atoms in the transition state ring by which the isomerization took place and the type of C–H bond being broken. We therefore assumed that the same transition state rings would have the same energy barriers for O₂QOOH species isomerization. That is, we have assumed that the transition state ring strain energy for each isomerization reaction should depend only on the size of the transition state ring and the atoms arranged in that transition state ring, i.e., this energy would be the same in an RO₂ reactant as in a O₂-QOOH reactant, with the only differences being the bond energy of the C–H bond that is broken.
3. Curran et al. [24] assumed that the C–H bonds for the C atom holding the OOH moiety in the initial RO₂ species have a bond energy 3 kcal/mol lower than the original value. Following this, in the revised mechanism, we assumed that the O₂QOOH radical isomerization reactions have the same rates as their logically similar RO₂ isomerizations, but with an activation energy 3 kcal/mol lower. The Nancy kinetic modeling group [25,26] uses the same logic but reduce this activation energy by only 2 kcal/mol. As further theoretical studies of reaction classes (C-22) and (C-23) become available, we will use them to replace the current rate expressions, but the present approximations are likely to be considerably better than past values, since they contain much more of the fundamental chemistry features of these reactions.

Uncertainties in the rates of the RO₂ isomerization reactions have been a concern for many years, and the inadequacy of the Pollard/Fish approach used previously was noted by Villano et al. [43]. Earlier studies by Baldwin et al. [54], Hughes et al. [55], and Robertson et al. [56] on the energy barriers for the multiple RO₂ isomerization pathways led to improved rates employed by Buda et al. [25] and discussed by Battin-Leclerc [26]. However, until experimentally measured chemical species concentrations specific to the low temperature reaction pathways were available in addition to the macroscopic data such as ignition delay times, the details of the low temperature reaction pathways and rates had little impact on kinetic models. The most important quantity that is required of low temperature kinetic models has been the ratio of chain branching to chain propagation within the NTC regime, since this parameter controls the rate of heat release in the NTC region and therefore the extent to which ignition advances as a result of low temperature chemistry [26,57]. This ratio can be provided accurately by rather crude kinetic mechanisms, explaining why global, lumped, or significantly reduced kinetic mechanisms can accurately reproduce complex ignition delay behavior for many hydrocarbon fuels, even with substantial contributions from low temperature reaction pathways. The present measurement of heptene isomers; however, could not be accurately simulated without more accurate low temperature kinetic rate parameters.

With the modified low temperature kinetic parameters, new computational results were generated for the pressure time histories and intermediate species. Figures 8 and 10 present the results

of the modified rate parameters. The computed results for n -C₇H₁₆ ignition showed significantly better results for the heptene isomers, as well as improved agreement for many other species with the experimental results. The computed ignition delay times with the improved low temperature kinetics were still faster than the experimental values. Specifically, the first stage ignition was about 4 ms and the second stage ignition was at about 10 ms, compared with the experimental values of 8.5 ms and 14.5 ms. The greatly improved predictions for 2-C₇H₁₄ and 3-C₇H₁₄ and smaller changes in other species predictions indicate that the new mechanism is significantly improved.

Of the 17 species measured experimentally, all of the concentrations computed from the kinetic model are in reasonable agreement with the measurements except the fuel and two aldehydes—propionaldehyde (C₂H₅CHO) and n -butyraldehyde (n -C₃H₇CHO). C₂H₅CHO and n -C₃H₇CHO are assumed to be produced in the new mechanism by a number of low temperature reactions that are quite speculative. Little attention has been given to C₂H₅CHO and n -C₃H₇CHO formation or consumption reactions and much more work would be needed to be more confident in their kinetics. We have therefore not included their computational predictions in the Fig. 8. The differences between the measured and computed consumption of n -C₇H₁₆ are more troublesome, especially since the amounts of n -C₇H₁₆ in both the present experiments and the experiments of Minetti et al. [13] disagree with the high extents of consumption predicted by the kinetic mechanism. Further analysis of these results will be needed to resolve this discrepancy, but it is interesting to note that even with this difference in fuel consumption, the model is able to predict the concentrations of so many intermediate species as seen in the rest of these species profiles.

5. Discussion

Using the present, upgraded kinetic reaction mechanism, n -C₇H₁₆ autoignition was simulated for the same constant volume condition using the pressure and mixture composition described earlier, but over a wider temperature range from 600 K to 800 K. The results were used to calculate the difference in values of τ_{ign} and τ_1 as a function of initial temperature, as well as the percentage of the n -C₇H₁₆ fuel consumed during the first stage of ignition. The results are plotted in Fig. 11. It is clear from this figure that the nature of the ignition changes profoundly as the initial temperature increases from 600 K to 800 K. For temperatures below 650 K, the ignition delay decreases at an increasing rate as the temperature increases, with the duration of the first stage of ignition forming a large portion of the overall ignition delay. As the temperature nears 700 K, where the present experiments were carried

out, the total ignition delay time reaches a local minimum and the interval between τ_1 and τ_{ign} increases. At 700 K, Fig. 11a shows an interval of about 6–7 ms between τ_1 and τ_{ign} events, essentially the same as the interval measured in the present experiments and in the final model simulations (see Fig. 10). This indicates that the major difference between the model predictions and the experiments is the time of the first stage of the ignition. Also, according to the model, the amount of fuel consumed at the first stage decreases as the initial temperature increases and as the duration of the second stage increases. Yet, the good agreement between the model and experimental concentrations of the species in Fig. 8 indicates that the detailed chemistry of the ignition is otherwise well reproduced by the kinetic model.

Two factors make kinetic simulations of n -C₇H₁₆ autoignition particularly challenging at temperatures near 700 K, in particular the complex transition taking place in the important reaction pathways from fuel consumption by H atom abstraction reactions with RO₂ and HO₂ radicals below 700 K to abstraction reactions with OH and H radicals at higher temperatures. Also at approximately 700 K, the amount of fuel consumed at the first stage starts to decrease significantly with increasing temperature, as seen in Fig. 11b. This complex temperature range will need further attention to clarify the details of the reaction rates and pathways, and our results indicate that many of the kinetic processes are quite sensitive to this relatively narrow temperature range.

The difficulty of predicting n -C₇H₁₆ consumption at 700 K has been seen frequently in kinetic modeling studies. In their experimental and kinetic modeling study of fuel consumption at 750 K during the compression stroke in a rapid compression machine, Cox et al. [14] calculated approximately 40% n -C₇H₁₆ fuel conversion in the first stage ignition while their experiments showed 10–20% conversion. In RCM experiments at 667 K, Minetti et al. [13] measured 20–30% n -C₇H₁₆ consumption in the first stage ignition and calculated 80% fuel consumption, as noted earlier. In a very recent jet-stirred reactor study that is similar in some regards to the present study, combining excellent experimental analysis over a wide range of reactor temperatures (500–1100 K) with detailed chemical kinetic modeling, Herbinet et al. [9] provided very good agreement between measured and computed species levels over the entire range of temperatures for nearly all of the 47 different species measured. Herbinet et al. [9] note in particular for n -C₇H₁₆ that their model “reproduces well the reactivity over the whole range of temperatures.” However, a close examination of the plotted results (Fig. 2 of [9]) shows that there is a difference between the computed and measured mole fractions of n -C₇H₁₆ of a factor of two at a reactor temperature of 700 K (computed mole fraction $\sim 2 \times 10^{-3}$, experimental mole fraction $\sim 4 \times 10^{-3}$), although the model reproduces the rather complex overall shape

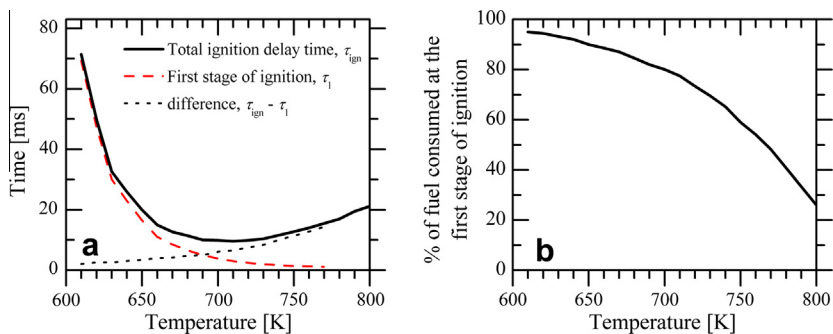


Fig. 11. The panel on the left shows the sensitivity of the new n -C₇H₁₆ mechanism predictions ($P_0 = 9$ atm, $\chi(n\text{-C}_7\text{H}_{16}) = 0.0134$, $\chi(\text{O}_2) = 0.149$, $\chi(\text{N}_2) = 0.2336$, and $\chi(\text{CO}_2) = 0.6040$) of the first and second stages of ignition to temperature. The panel on the right shows that the predicted amount of fuel consumed at the first stage of ignition decreases significantly above 700 K.

of the n -C₇H₁₆ mole fraction variation over the full temperature range. It is interesting to note that Herbinet et al. [9] predicted an excess consumption of n -C₇H₁₆ of about the same magnitude as the present work at about 700 K, even after including a similar family of low temperature RO₂ isomerization and other reaction rates as those in our upgraded kinetic reaction mechanism, leaving a good explanation for the over-prediction of fuel consumption elusive.

The changes made to produce the new n -C₇H₁₆ kinetic mechanism involved improved rates of reactions forming RO₂ and their subsequent isomerization and decomposition reactions; the subsequent addition of O₂ to the RO₂ isomerization products QOOH; and the isomerization of the O₂QOOH species to produce OH and ketohydroperoxide intermediates. The opportunity to upgrade the current mechanism with these new rate values based on theory rather than estimations offers some new interesting insights into the overall kinetic reaction mechanism. Villano et al. [43] conducted important calculations to suggest an increase in the fraction of RO₂ radicals that react via 1–5 and 1–6 isomerizations, that is, via 6-atom (•CCCOOH) and 7-atom (•CCCCOOH) transition state rings, as opposed to 1–4 or 1–7 isomerizations. They noted that recent electronic structure calculations by Davis and Francisco [53], not possible previously when the original kinetic mechanisms for fuels such as n -C₇H₁₆ had been developed, showed that the incorporation of the O atoms into transition state rings has a considerable impact on the energetics for these isomerizations not seen in alkyl radical isomerization reactions. Specifically, the 1–5 and 1–6 isomerization rates are faster by factors of nearly 30 for RO₂ radicals than the previous rates based on analogous alkyl radical isomerizations, making RO₂ isomerizations the dominant reaction pathways for the low temperature kinetics. The 1–5 reaction leads primarily to QOOH products that favor O₂ addition and eventually chain branching, while the 1–6 reaction leads to QOOH cyclization and chain propagation. The dominance of these two reaction pathways simplifies the balance between branching and propagation that determines the amount of NTC behavior observed in both the experiments and the kinetic model. Such information could be valuable in developing reduced kinetic models for ignition of all types of hydrocarbon fuels.

The reduced fraction of RO₂ reaction via 1–4 isomerization in the revised mechanism leads to less heptene production, further reducing the excess heptene production observed in the simulations prior to the mechanism upgrade and seen as the dashed lines in Fig. 8b and c. The enhanced fraction of the 1–6 isomerization leads to the relatively high levels of large cyclic ethers in the intermediate species composition. In Section 3.2 above, we attributed most of the missing carbon in the experimental carbon balance to large C₇ cyclic ethers, and the relatively high rates of the 1–6 isomerizations lead preferentially to large amounts of C₇ cyclic ethers with 4 C atoms and one O atom. From the detailed output from the upgraded kinetic model, we find that most (i.e., about 75%) of the missing carbon is found in the form of C₇H₁₄O₂–5. This is the cyclic ether produced via H atom abstraction at either the 2-site or the 3-site in n -C₇H₁₆ (both of them being secondary sites with weakly bonded H atoms), followed by addition of O₂ to the radical site to produce either C₇H₁₅O₂–2 or C₇H₁₅O₂–3 and then the 1–6 isomerization reaction to produce the QOOH species C₇H₁₄OOH₂–5 or C₇H₁₄OOH₃–6, both of which then decompose to produce the same cyclic ether, C₇H₁₄O₂–5. Each step of these sequences follows an energetically favored path, from the initial H atom abstraction from the fuel at secondary C–H sites to the energetically favored 1–6 isomerization reaction, and then the cyclization reaction to produce the –C–C–C–C–O– ring. We note that the cyclization reaction of the 1–6 isomerization product has a rate at 700 K that is 20 times faster than the similar cyclization reaction of the 1–5 isomerization reaction product at the same temperature. Under the

present conditions, the cyclization reaction rate is faster than the addition rate of molecular oxygen to either the QOOH species C₇H₁₄OOH₂–5 or C₇H₁₄OOH₃–6, so the dominant reaction pathway for products of 1–6 isomerization is cyclization. Cyclization also yields a single OH product, so cyclization is effectively a chain propagation reaction pathway. In contrast, the product of 1–5 isomerization is much more likely to react via addition of molecular oxygen and subsequent chain branching, since its rate of cyclization is much smaller than the cyclization reactions of the 1–6 isomerization products. For these reasons, the 1–5 RO₂ isomerization reaction is generally regarded as a chain branching pathway while the 1–6 RO₂ isomerization reaction is generally regarded as a chain propagation reaction pathway. Since the 1–4 and 1–7 RO₂ isomerization reaction rates are significantly smaller than the 1–5 and 1–6 isomerization rates, the major factor influencing the relative amounts of chain propagation and chain branching in the present mechanism are the relative rates of the 1–5 and 1–6 isomerization rates.

A second observation provided by the new, more theory-based low temperature kinetic rates is that the production of heptenes is significantly reduced relative to the previous mechanism because the new rates of the concerted elimination reactions, reaction class (C-12b), are slower than in the previous mechanism. Reaction class (C-12b) produces a heptene species and HO₂ and is thus formally a chain propagation step, but it actually limits the overall rate of ignition more strongly than other chain propagation reactions because HO₂ is less reactive than most other radicals. In the Mehl et al. mechanism [28], the rate of reaction class (C-12b) had to be reduced in order to retard the time of ignition, but that led to extremely high computed levels of heptenes. A firm establishment of the rates of these reactions greatly reduced uncertainties in many of the other low temperature reactions and reaction pathways.

6. Conclusions

Speciation data presented in this work provide important new insights into the intermediates formed during the ignition delay of n -heptane. Computational results using an established chemical kinetic mechanism that has been widely used for many years [24,28] for the concentrations of several important intermediate species such as heptenes initially showed poor agreement with the speciation data (obtained at an effective pressure of 9 atm and an effective temperature of 700 K) from the current work. A refined chemical kinetic reaction mechanism that includes low temperature rate expressions taken from newly published theoretical studies by Villano et al. [43,51] predicts ignition delay times and intermediate species concentrations that are in much better agreement with experimental data; however, the mechanism predicts higher consumption of n -heptane at the first stage of ignition and predicts more heat release at the first stage of ignition than observed experimentally.

While previous n -heptane ignition studies have provided remarkable understanding of the effects of a broad range of temperatures and pressures on ignition delay times, few experimental studies have interrogated the reaction pathways proposed and represented in chemical kinetic mechanisms in the form of intermediate chemical species concentrations. The present study provides new measurements of intermediate species levels during n -heptane ignition in the low-temperature regime that could not be reproduced using currently available kinetic mechanisms. The discrepancies are largely due to poorly known rates of alkylperoxy radical reactions and reactions of other species produced via alkylperoxy isomerization reactions. New theory results of Villano et al. [43,51] reduced these uncertainties, and improvements in the mechanism made possible by these new studies were key to improving the agreement between computed and measured

intermediate species concentrations. Previous rates for alkylperoxy and alkylperoxy isomerization reactions had been based primarily on well-intentioned estimates, so the new rate information, based on established theoretical principles, are a significant upgrade to current kinetic models. While improvements have been made in this study for kinetics of *n*-heptane oxidation, considerable uncertainties remain. New studies of these reactions, particularly including experiments that address oxidation reactions and rates in the low temperature range are critical to making further mechanism refinements in this very important temperature regime.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.combustflame.2013.06.029>.

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